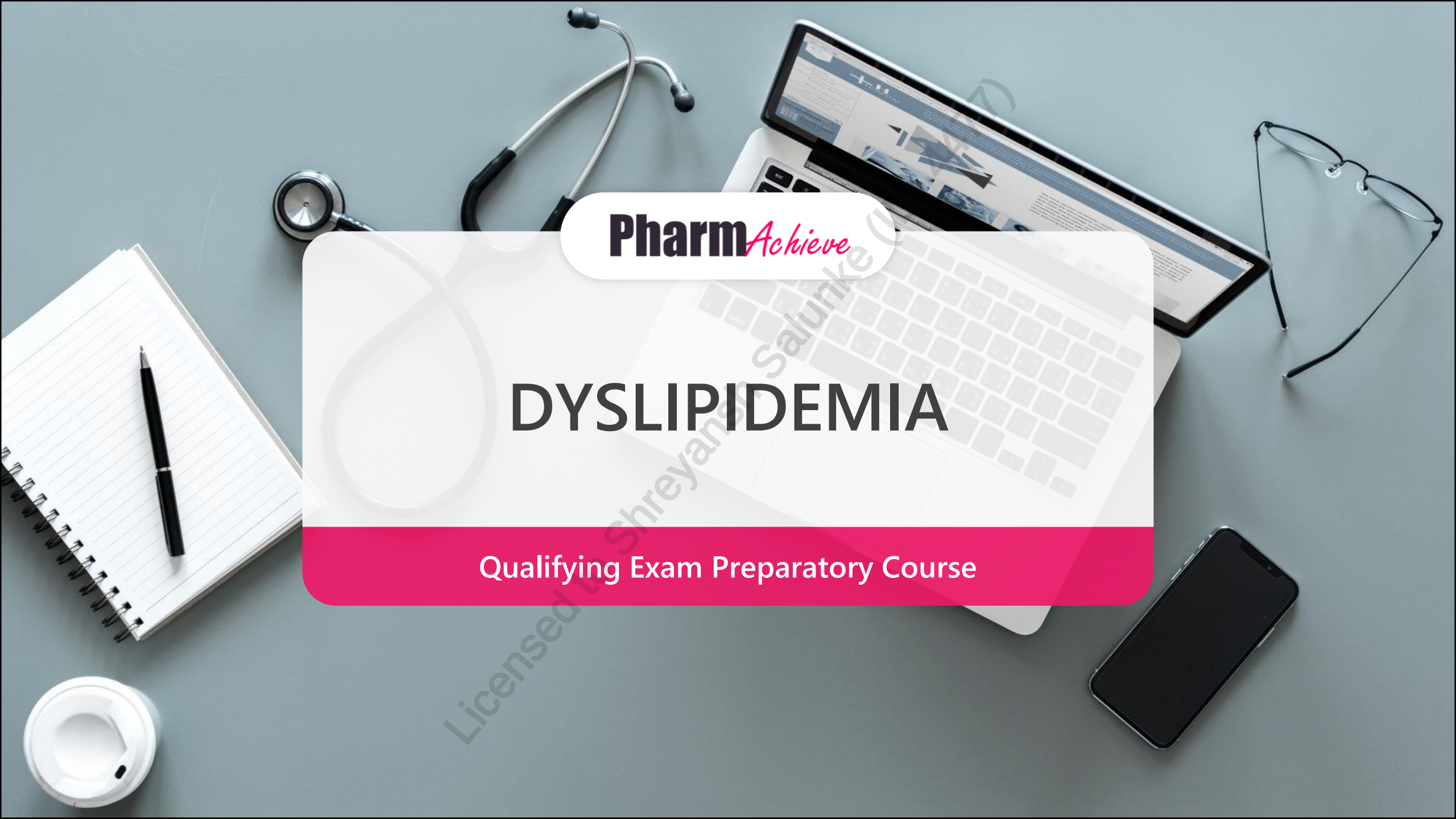




Pharm*Achieve*

DYSLIPIDEMIA

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

ApoB Apolipoprotein B

AS Ankylosing Spondylitis

BMI Body Mass Index

CAD Coronary Artery Disease

CK Creatine Kinase

CKD Chronic Kidney Disease

COPD Chronic Obstructive Pulmonary Disease

CVD Cardiovascular Disease

CVA Cerebrovascular Accident

CVD Cardiovascular Disease

HDL High-Density Lipoprotein

HTN Hypertension

IBD Inflammatory Bowel Disease

IFG Impaired Fasting Glucose

LDL Low-Density Lipoprotein

LFT Liver Function Test

Lp(a) Lipoprotein (a)

MI Myocardial Infarction

PAD Peripheral Artery Disease

PsA Psoriatic Arthritis

PVD Peripheral Vascular Disease

RA Rheumatoid Arthritis

SLE Systemic Lupus Erythematosus

TB Tuberculosis

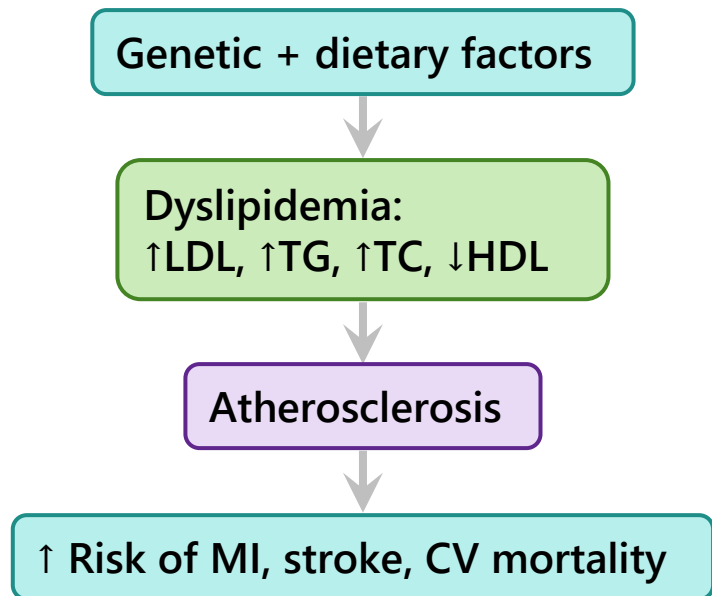
TC Total Cholesterol

TG Triglycerides

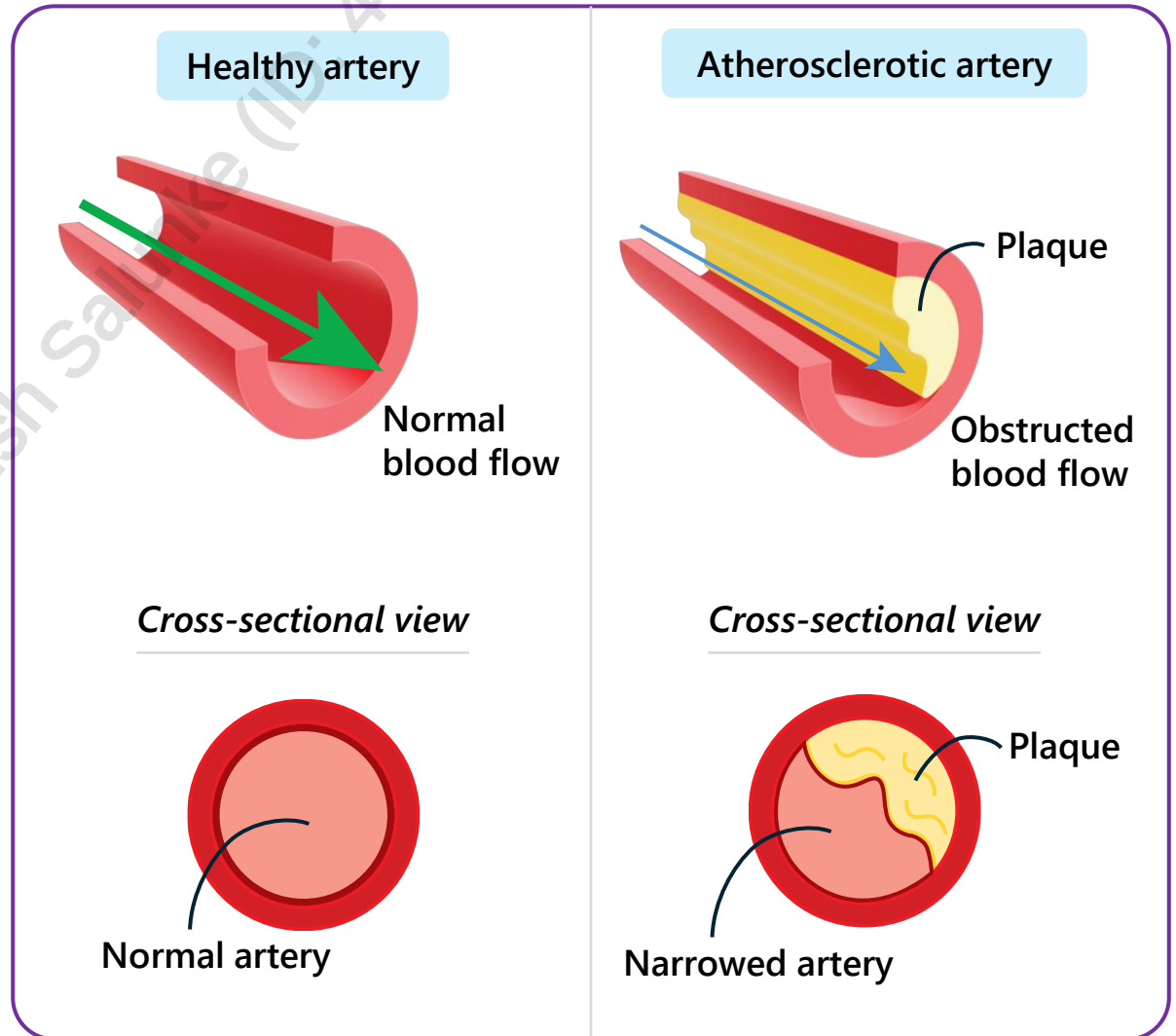
URTI Upper Respiratory Tract Infection

PATHOPHYSIOLOGY

LDL cholesterol levels are directly linked to the development of atherosclerosis and its reduction is directly linked to reduction in cardiovascular disease



While LDL-C is the main therapeutic target for ASCVD prevention, strong evidence continuously emerges favouring apoB as a more accurate indicator of CV risk



SCREENING

All men and women ≥ 40 years of age or post-menopausal women

All patients (regardless of age) with any of the following:

- Clinical evidence of atherosclerosis
- Abdominal aortic aneurysm
- Diabetes mellitus
- Arterial hypertension
- Smoking tobacco
- Stigmata of dyslipidemia
- Family history of dyslipidemia or premature CVD
- Chronic kidney disease $\text{eGFR} \leq 60 \text{ mL/min/1.73m}^2$ or $\text{ACR} \geq 3 \text{ mg/mmol}$ for ≥ 3 months
- Obesity ($\text{BMI} \geq 30 \text{ kg/m}^2$)
- Inflammatory disease (RA, SLE, PsA, AS, IBD)
- HIV
- Erectile dysfunction
- COPD
- History of hypertensive disorder of pregnancy

****Consider earlier screening in ethnic groups at higher risk such as South Asian or Indigenous individuals**

SECONDARY CAUSES OF HYPERLIPIDEMIA

Drug causes

- Thiazide diuretics
- Azole antifungals
- Hormone replacement therapy
- Oral contraceptives
- Corticosteroids
- Isotretinoin
- Mirtazapine
- Atypical antipsychotics
- Cyclosporine
- Amiodarone
- HIV medications (e.g. protease inhibitors)

Other causes

- Metabolic syndrome (general term used to define a cluster of conditions that occur together which ultimately increases the risk of heart attack, stroke, and diabetes)
- High alcohol intake
- Pregnancy
- CKD or liver disease
- Hypothyroidism
- Biliary obstruction
- Obesity

RISK FACTORS FOR METABOLIC SYNDROME

- Waist circumference >102 cm in men or >88 cm in women
- Triglycerides ≥ 1.7 mmol/L
- HDL <1 mmol/L in men or <1.3 mmol/L in women
- Systolic BP ≥ 130 mmHg
- Diastolic BP ≥ 85 mmHg
- Fasting blood glucose ≥ 5.6 mmol/L

CVD RISK FACTORS

- Men >55 years old, Women >65 years old
- Smoking
- Diabetes Mellitus
- Hypertension
- Increased LDL
- Decreased HDL
- Family history of pre-mature CVD (first-degree male relative <55 years or first-degree female relative <65 years)
- Abdominal obesity
- Sedentary lifestyle
- Increased Triglycerides
- Excess alcohol consumption
- Poor diet

FRAMINGHAM RISK SCORE

Primary prevention CVD tool for estimating a patient's 10-year risk of developing CVD

Enter the following information in the online calculator to determine risk:

- Age
- Sex
- Total cholesterol (mmol/L)
- HDL cholesterol (mmol/L)
- Smoking status
- Diabetes co-morbidity
- Systolic blood pressure (mmHg)
- Is the patient being treated for high blood pressure?
- Is there a family history of premature CVD?

Limitations

- Lack of generalizability: Based on Caucasian middle-class male Americans – questionable reliability in patients of other ethnic origins
- Cannot be used in patients with CKD
- Does not accurately predict risk for the young, elderly (> 75 years), and female patients
- Predicts risk of 1st event, not subsequent events
- Doesn't consider premature family history of CVD

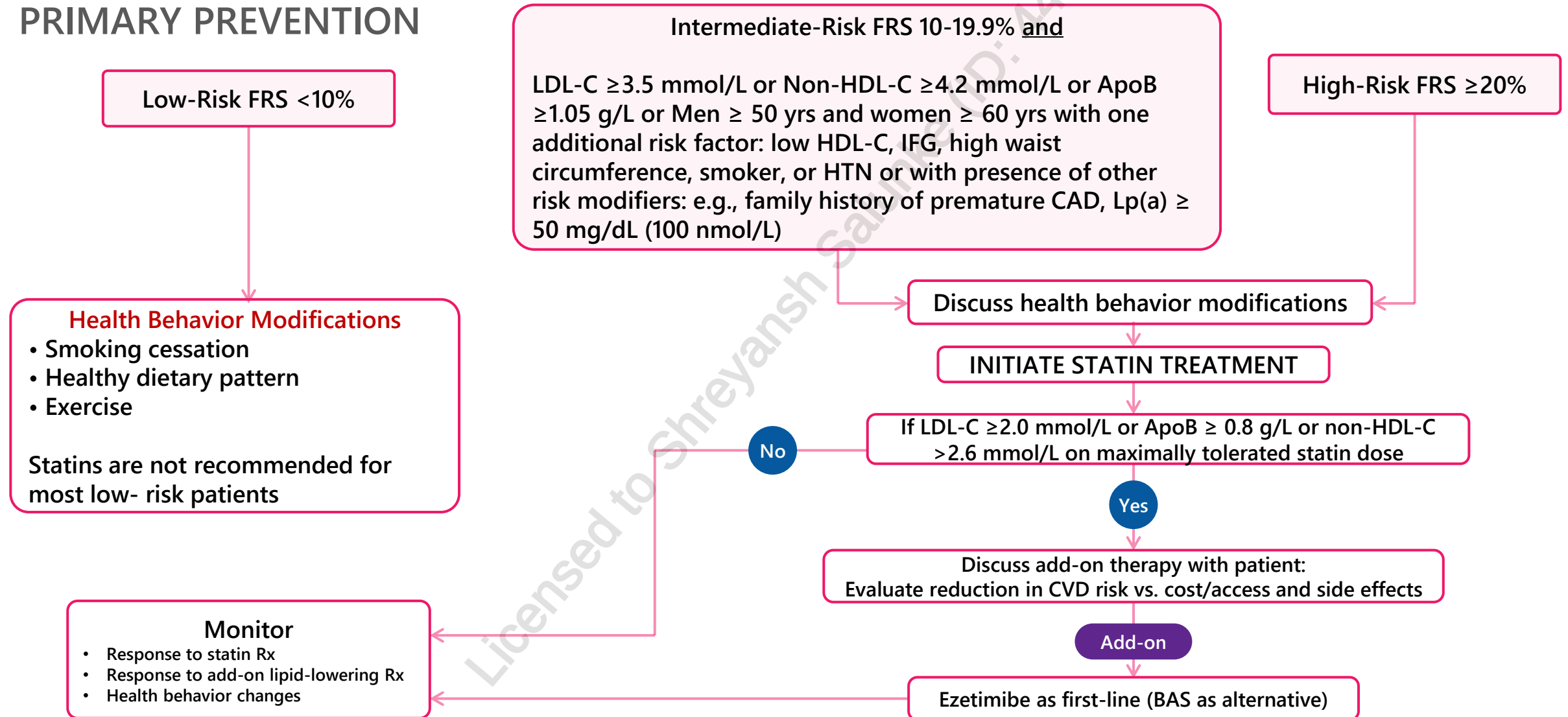


Note: Risk score is doubled in the presence of premature CVD family history

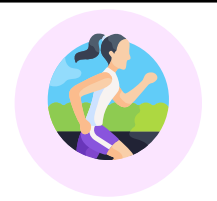
RISK STRATIFICATION

Risk should be reassessed every 5 years if FRS < 5% and annually if FRS ≥ 5%, unless there is a statin-indicated condition

PRIMARY PREVENTION



NON-PHARMACOLOGICAL MEASURES



CORNERSTONE OF CHRONIC DISEASE PREVENTION

Smoking Cessation

The most important health behavior intervention for the prevention of CVD (linear and dose-dependent association between the number of cigarettes smoked per day and CVD risk)

Consuming a healthy diet

- Mediterranean dietary pattern (strong recommendation)
- DASH dietary pattern (conditional recommendation)
- Plant-based dietary pattern
- Diet rich in fruits, vegetables, nuts, legumes, olive oil, total fibre, whole grains
- Increase fiber to >30 g/day, reduce cholesterol to <300 mg/day, and reduce sodium to <2 g/day

Engaging in regular physical activity

- At least 150 minutes of moderate-to-vigorous aerobic activity per week
- Beneficial to add muscle- and bone-strengthening activities at least 2 days per week

Maintaining a healthy body weight

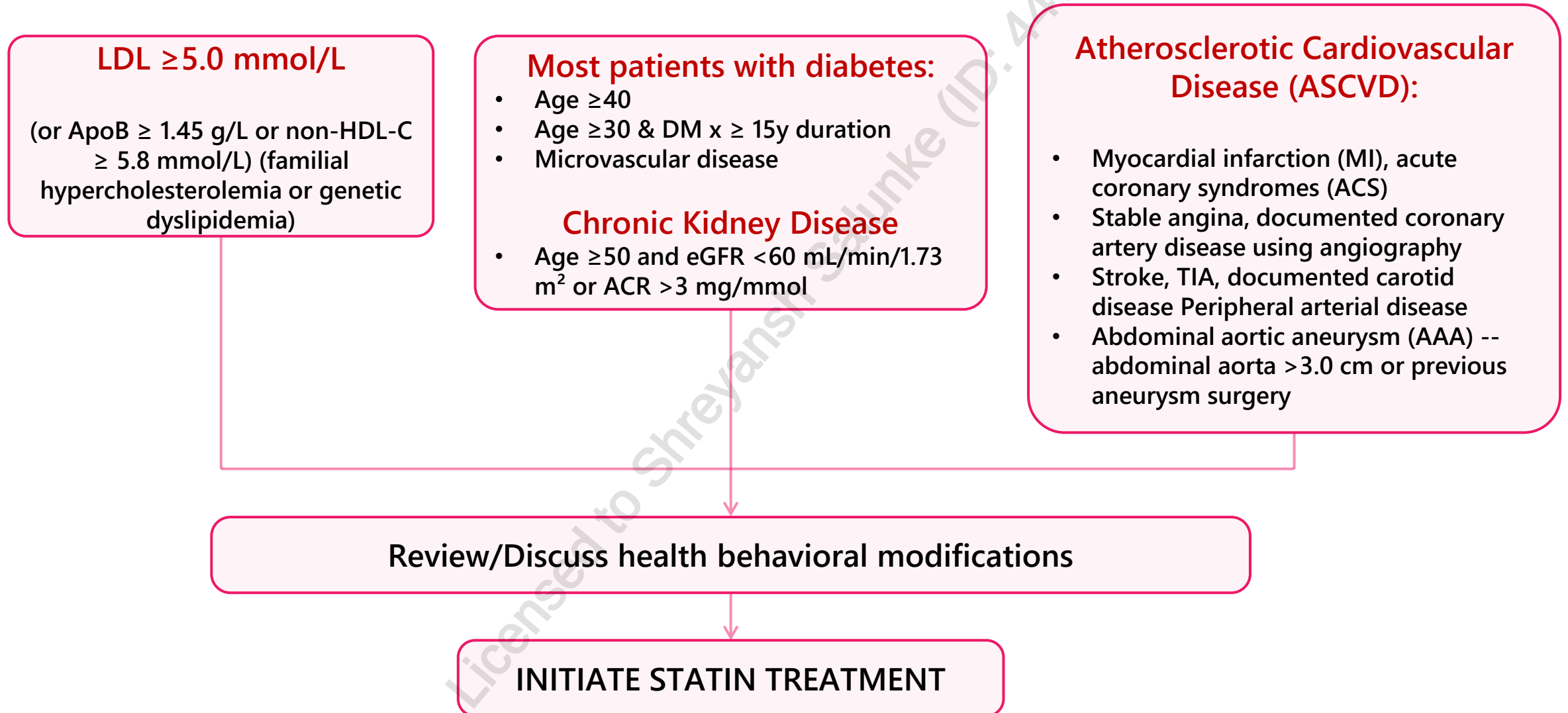
Limit alcohol intake

Ensuring sufficient duration of sleep

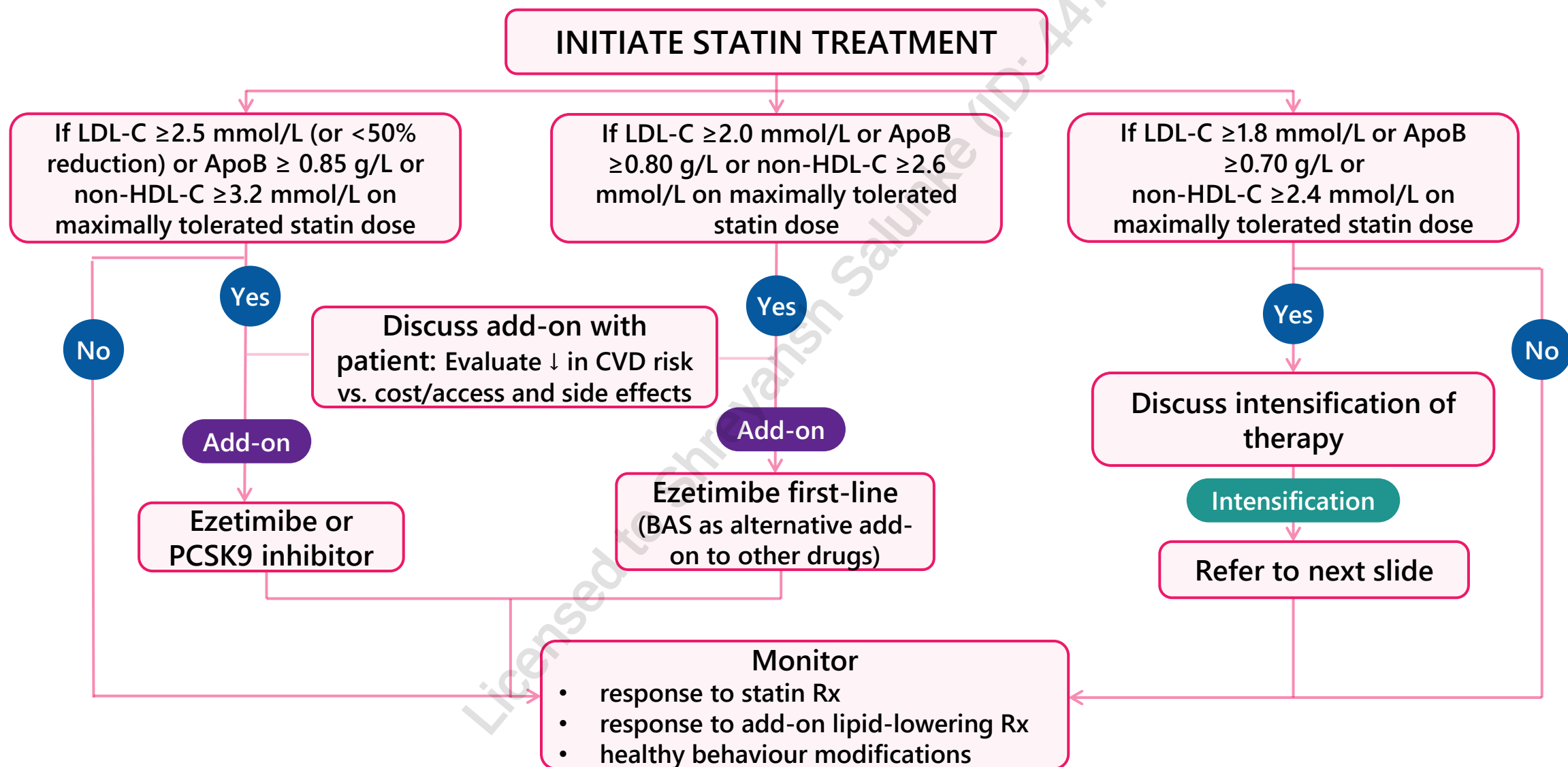
Stress management

Note: Omega-3 polyunsaturated fatty acid supplements are not recommended due to lack of CV benefit

STATIN INDICATED CONDITIONS



STATIN INDICATED CONDITIONS (Continued)



PATIENTS WITH ATHEROSCLEROTIC CVD

Patients with ASCVD on maximally tolerated statin dose

If LDL-C is ≥ 1.8 mmol/L or
if ApoB ≥ 0.70 g/L or if non-HDL-C ≥ 2.4 mmol/L

LDL-C 1.8-2.2 mmol/L or
ApoB 0.70-0.80 g/L or non-
HDL-C 2.4-2.9 mmol/L

Consider
ezetimibe +/- PCSK9
inhibitor*

LDL-C > 2.2 mmol/L or
ApoB > 0.80 g/L or
non-HDL-C > 2.9 mmol/L or
high PCSK9i benefit patient*

Consider
PCSK9 inhibitor* +/-
ezetimibe

If TG is ≥ 1.5 to 5.6 mmol/L

Consider
Icosapent ethyl 2000 mg BID†

† May also be considered for patients without ASCVD but with DM requiring medication treatment in patient ≥ 50 years of age. and 1 additional CV risk factor:

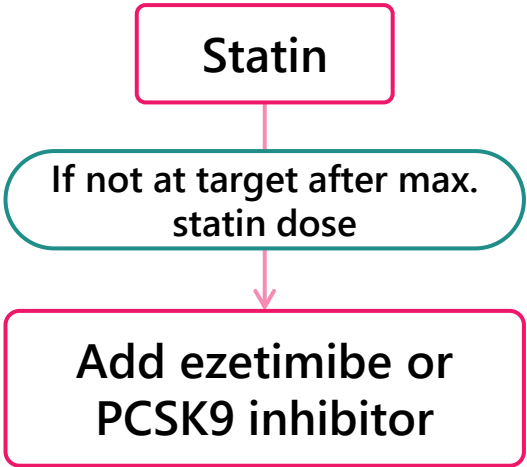
- men ≥ 55 y and women ≥ 65 y
- cigarette smoker or stopped smoking within 3 months
- hypertension (≥ 140 mmHg systolic OR ≥ 90 mmHg diastolic) or on BP medications
- HDL-C ≤ 1.04 mmol/L for men or ≤ 1.3 mmol/L for women
- hsCRP > 3.0 mg/L
- Renal dysfunction: eGFR > 30 and < 60 mL/min
- Retinopathy
- Micro- or macroalbuminuria
- ABI < 0.9 without symptoms of intermittent claudication

* PCSK9i preferred over ezetimibe if the patient has DM, metabolic syndrome, polyvascular disease, symptomatic PAD, recurrent MI, MI in past 2 years, previous CABG, LDL ≥ 2.6 mmol/L, heterozygous familial hypercholesterolemia

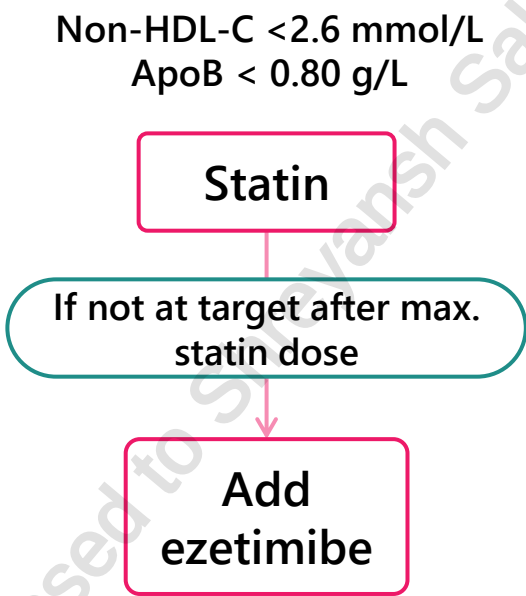
SUMMARY OF STATIN

INDICATED TREATMENT AND TARGETS

LDL ≥ 5 :
(target LDL < 2.5)

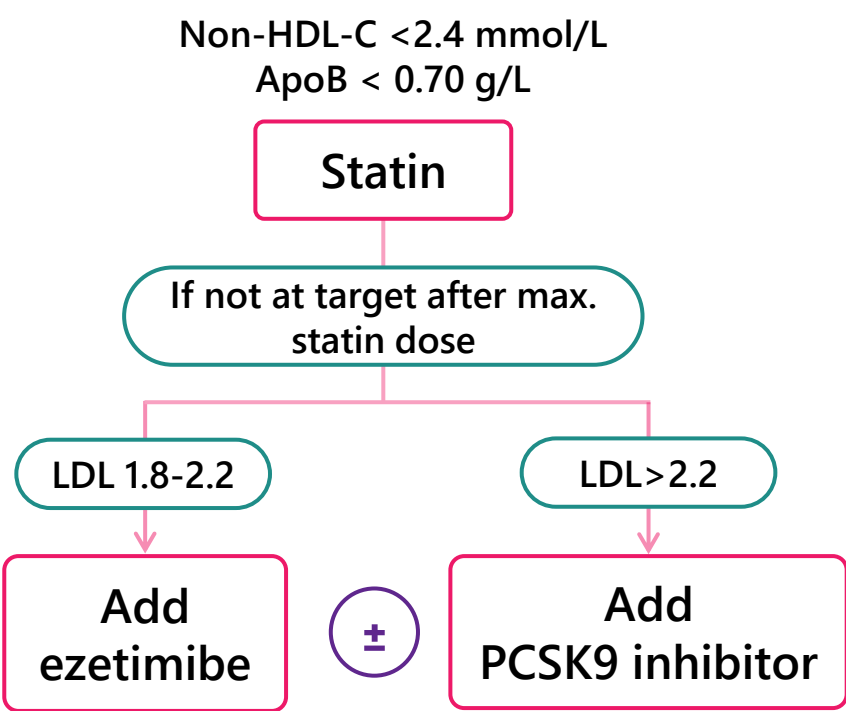


CKD* or DM:**
(target LDL < 2)



* age ≥ 50 and eGFR < 60 mL/min/1.73 m² or ACR > 3 mg/mmol
** age ≥ 40 or ≥ 30 and DM for ≥ 15 years or microvascular disease

ASCVD:
(target LDL < 1.8)



PRIMARY AND SECONDARY LIPOPROTEIN DETERMINANTS

FYI

**Chylomicron
Remnants**

VLDL
Very low-density
lipoprotein

IDL
Intermediate-density
lipoprotein

Lp(a)
Lipoprotein

LDL
Low-density
lipoprotein

HDL
High-density
lipoprotein

Non-HDL-C = Total Cholesterol-HDL-C
Atherogenic
Apo B-containing
Lipoproteins

LDL-C: Calculated from standard
Lipid Profile
ApoB: Measured separately

HOW TO SCREEN

For all:

- History and physical examination
- Standard lipid panel (TC, LDL-C, HDL-C, TG)
- Non-HDL-C (will be calculated from profile)
- Glucose
- eGFR

Optional:

- ApoB
- Urine albumin: creatinine ratio (if eGFR <60 mL/min/1.73m², hypertension or diabetes)

NON-FASTING LIPID TESTING IS ACCEPTABLE

✓ For any patient with triglycerides > 1.5 mmol/L, non-HDL-C or ApoB be used instead of LDL-C as the preferred lipid parameter for screening and treatment target

✓ ApoB and non-HDL-C are the most reliable lipid predictors of CVD

PHARMACOLOGICAL MEASURES

Statins (1st line): ↓CVD risk and mortality

- Rosuvastatin: 5-40mg daily
- Atorvastatin: 10-80mg daily
- Simvastatin: 10-80mg QHS
- Lovastatin: 20-80mg dinner
- Pravastatin: 10-40mg QHS
- Fluvastatin/Fluvastatin XL: 20-80mg daily-BID (depending on formulation)



High

Efficacy

Low

PCSK9 inhibitors: alirocumab, evolocumab [monoclonal antibodies administered SC], and inclisiran sodium [siRNA PCSK9i]: ↓LDL; Most common adverse effect is injection site reactions. No known drug interactions.

Fibrates (fenofibrate): Most potent for ↓TG (especially if TG > 10.0 mmol/L and no response to non-pharms)

- Subgroup analysis demonstrated potential benefit in patients with elevated TG and low HDL-C levels

Ezetimibe: Add-on agents if target LDL levels are not met with maximally tolerated statin monotherapy

Bile acid sequestrants (colestipol, cholestyramine): take with water and fibre, separate from other medications.

Ethyl Eicosapentaenoic Acid (EPA) (Icosapent Ethyl): If TG is ≥ 1.5 to 5.6mmol/L

- May also consider for patients without ASCVD but with DM requiring medication treatment in patients ≥ 50 years of age and ≥ 1 additional CV risk factor
- May increase risk of bleeding if taken concomitantly with blood thinners

SUMMARY OF EFFICACY

Class/drug name	LDL	HDL	TG	Mortality benefits
Statins	↓ 20-60%	↑ 5-15%	↓ 7-30%	++
Resins	↓ 15-30%	↑ 3-10%	POSSIBLE INCREASE	Cholestyramine: +
Fibrates	↓ 5-20%	↑ 10-35%	↓ 20-50%	
Ezetimibe	↓ 14-20%	↑ 1%	↓ 7-9%	ACS: + (in combo with statin)
Niacin	↓ 5-25%	↑ 15-35%	↓ 20-50%	
PCSK9 inhibitors	↓ 50-70%			+ (In combo with statin)

Safety notes:

- Statins, ezetimibe, and fibrates have all been associated with myalgias. **Bile acid sequestrants such as cholestyramine are not associated with myalgias.**
- Gemfibrozil (fibrate) and niacin increase the risk of myopathy when used concurrently with statins

STATINS- SIDE EFFECTS

- Myalgia: muscle pain/weakness **without CK rise**
- Myositis: muscle pain/weakness + **↑CK**
- Rhabdomyolysis: **CK > 10 x UNL, ↑SCr, renal failure**
- There have been cases of confusion and memory loss associated with high-dose statin therapy

Risk factors: elderly, female, small body frame, renal insufficiency, diabetes, hypothyroidism, combination lipid-lowering agents, drug interactions



****Pravastatin and rosuvastatin are least likely to cause muscle pains, but there are reports of rhabdomyolysis with 40mg doses of rosuvastatin****

STATINS- MANAGEMENT OF SIDE EFFECTS

Management of myalgia: may continue or stop statin depending on severity of pain (if statin stopped, advised to wait until pain resolves before restarting)

- **Re-initiation once patient asymptomatic:** May consider lowering the dose OR switching to another statin OR alternate day dosing
-

Management of myositis: stop statin temporarily

- **Re-initiation once CK levels normal and patient asymptomatic:** may consider lowering the dose OR switching to another statin OR alternate day dosing
-

Management of rhabdomyolysis: stop statin and assess renal function

- If patient has high CV risk, alternatives are recommended e.g. PCSK9 inhibitors
- **Re-initiation is considered:** may use a low dose of a different statin, if mild case

STATINS- CLINICAL PEARLS

METABOLISM

- Simvastatin, atorvastatin, and lovastatin are primarily metabolized by CYP3A4
- Rosuvastatin and fluvastatin are primarily metabolized by CYP2C9
- Pravastatin is not metabolized by CYP enzymes and therefore has the least amount of drug interactions

CONTRAINDICATIONS

- Active liver disease, pregnancy and breastfeeding

CLINICALLY SIGNIFICANT DRUG INTERACTIONS

- Do not combine statins with cyclosporine or gemfibrozil due to increased concentration of statins when used together

SIDE EFFECTS OF OTHER AGENTS

Ezetimibe*

- Back pain
- Arthralgia
- Diarrhea
- Abdominal pain
- Flatulence
- Headache

- *Not recommended in moderate-severe hepatic dysfunction
- *Caution with concomitant ezetimibe and cyclosporin use, especially in severe renal dysfunction
- *Cholestyramine reduces the absorption of ezetimibe

PCSK9 inhibitors

- Injection-site reactions
- Increased risk of infections (e.g. URTI)



Monoclonal antibodies, \$\$\$

PREGNANCY/HEPATIC IMPAIRMENT CONSIDERATIONS

PREGNANCY CONSIDERATIONS

- Statins, fibrates, ezetimibe, and PCSK9 inhibitors **should not be used** during pregnancy or while breastfeeding
- **Resins**, such as cholestyramine, may be considered throughout pregnancy and while breastfeeding with physician approval

HEPATIC IMPAIRMENT CONSIDERATIONS

- **Cholestyramine** is not absorbed from the gastrointestinal tract and therefore no dosage adjustment is required for any degree of hepatic impairment

MONITORING PARAMETERS

Efficacy Endpoints		
Parameter	Degree of Change	Timeframe
LDL, non-HDL, ApoB	At target (varies based on patient's conditions)	4-6 weeks

Safety Endpoints		
Parameter	Degree of Change	Timeframe
ADR from medication therapy e.g. myalgias, GI, injection-side reactions	Prevent or Minimize	Duration of therapy
Serum transaminases e.g. ALT	None – minimal	At least once at 3 months
CK levels	None – minimal	Do not check unless patient is symptomatic

TAKEAWAY

- **Low-risk** (FRS <10%) can generally be managed by health behavior modifications
- **Intermediate risk** (FRS 10%-19.9% plus additional risk factors) and **High-risk** (FRS $\geq 20\%$) necessitate statin initiation. Ezetimibe is added if LDL ≥ 2 mmol/L, or ApoB ≥ 0.8 g/L or non-HDL > 2.6 mmol/L
- Non-pharmacological interventions are the cornerstone of chronic disease prevention
- **Statin-indicated conditions:**
 1. **LDL ≥ 5 mmol/L** – If LDL still ≥ 2.5 mmol/L or ApoB ≥ 0.85 g/L or non-HDL ≥ 3.2 mmol/L $\rightarrow$ add ezetimibe or PCSK9 inhibitor
 2. **Diabetes (age ≥ 40 , or ≥ 30 and DM for ≥ 15 years or microvascular disease)** – if LDL ≥ 2 mmol/L, ApoB ≥ 0.8 g/L or non-HDL ≥ 2.6 mmol/L $\rightarrow$ add ezetimibe
 3. **CKD (age ≥ 50 and eGFR < 60 mL/min/1.73 m² or ACR > 3 mg/mmol)** - if LDL ≥ 2 mmol/L, ApoB ≥ 0.8 g/L or non-HDL ≥ 2.6 mmol/L $\rightarrow$ add ezetimibe
 4. **ASCVD** – if LDL 1.8 to 2.2 mmol/L $\rightarrow$ add ezetimibe ($\pm$ PCSK9 inhibitor). If LDL > 2.2 mmol/L $\rightarrow$ add PCSK9 inhibitor ($\pm$ ezetimibe). If TG ≥ 1.5 -5.6 mmol/L, add icosapent ethyl

CASE PRACTICE

TC is a 54-year-old male who presents with concerns regarding his cardiovascular risk, following a friend's recent myocardial infarction. During a routine annual physical examination with his general practitioner, he was informed of elevated low-density lipoprotein (LDL) cholesterol and subsequently received a new prescription. His medical history includes hypertension, well-controlled schizophrenia, gout, and depression. His current medication regimen consists of ramipril 5 mg orally once daily, risperidone 2 mg orally at bedtime, allopurinol 300 mg orally once daily, mirtazapine 7.5 mg orally at bedtime, acetaminophen 500 mg orally as needed for headaches, and vitamin B complex one tablet orally once daily. Social history indicates no tobacco use, consumption of approximately two alcoholic drinks daily, and irregular exercise. He reports no known drug allergies. TC's BMI is 24 Kg/m².

Vital signs: BP= 135/76 mmHg, HR=72 bpm

Lab values: Total cholesterol= 5.8 mmol/L, LDL-C = 3.1 mmol/L HDL-C = 1.2 mmol/L,
TG= 2.1 mmol/L, SCr = 89 umol/L

1. Which of his prescriptions is most likely to contribute to hyperlipidemia?
 - a) Ramipril
 - b) Allopurinol
 - c) Risperidone
 - d) Acetaminophen



CASE

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2. Which risk factor does Mr. TC have that would qualify him for primary prevention if he had an intermediate (10-19%) Framingham risk score?
- a) Hypertension
 - b) Alcohol intake
 - c) BMI
 - d) LDL-C



CASE

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TG= 2.1 mmol/L, SCr = 89 umol/L

3. Mr. TC is prescribed simvastatin 10 mg po QHS; what should be his drug therapy target?
- a) LDL-C < 2.0 mmol/L
 - b) Non-HDL-C < 2.0 mmol/L
 - c) LDL-C < 1.6 mmol/L
 - d) Apo-B < 0.4 g/L



CASE

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TG= 2.1 mmol/L, SCr = 89 umol/L

4. What counseling points should be reviewed with Mr. TC's regarding his simvastatin Rx?
- a) Take at night, avoid orange juice, monitor for muscle pain
 - b) Take at night, avoid grapefruit juice, encourage dietary changes and exercise
 - c) Take anytime of day and have liver function checked frequently
 - d) Take with food, avoid orange juice, decrease alcohol intake



REFERENCES

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4. Behbodikhah, J., Ahmed, S., Elyasi, A., Kasselman, L. J., De Leon, J., Glass, A. D., & Reiss, A. B. (2021). Apolipoprotein B and Cardiovascular Disease: Biomarker and Potential Therapeutic Target. *Metabolites*, 11(10), 690.

CHANGE LOG

March 2025

- Slide 15: changed "21 additional risk factor" to "1 additional risk factor" for icosapent

July 2025

- Updated legend
- Slide 10: Simplified and re-arranged information
- Slides 12, 16: Removed for redundancy
- Slide 13 & 14: simplified information
- Slides 17: Combined and consolidated information
- Slide 20: Consolidated and added information
- Slide 25: Reworded case study

August 2025

- Slides 7: Removed FYI

September 2025

- Formatting changes made throughout
- Slide 3: Added URTI
- Slide 15: Changed ">40 or >30 & DM for > 15 yrs" to all '≥' and added CKD patients
- Slide 22: Added URTI as an example of infections
- Slide 24: Elaborated on Takeaway

January 2026

- Minor formatting edits made throughout
- Slide 2: Updated NAPRA competencies slide
- Slide 9: Changed 'gender' to 'sex'
- Slide 10: Added screening timeframe for primary prevention. Added that statin therapy is not recommended for most low-risk patients
- Slide 11: Added note on omega-3 fatty acids
- Slide 21: Updated drug interactions
- Slide 24: Added monitoring parameters